

BudgetWise – Personal Finance Data Analysis

Complete Project Guide – Final Updated Version

1. Project Overview

BudgetWise is an Excel-based personal finance data analysis project.

The project starts with raw transaction data and follows a complete process of **data cleaning, quality checking, outlier review, data analysis, Power Pivot/DAX, and dashboard creation**.

The final dashboard helps understand financial transactions, income, categories, payment methods, and locations.

2. Project Workflow

Raw Data → Unique Value Analysis → Lookup Tables → Data Cleaning → QA Checks → Clean_Data → Outlier Review → PivotTables → Power Pivot / DAX → Dashboard

3. Workbook Sheets

The workbook contains the following main sheets:

- **Raw Data**
- **Lookup**
- **Cleaning**
- **Clean_Data**
- **QA_Check**
- **Outliers**
- **PivotTable sheets**
- **Dashboard**

4. Data Cleaning Process

The following steps were performed:

1. The original **Raw Data** was kept unchanged.
2. Unique values were checked for Category, Payment Mode, and Location.
3. Lookup tables were created for:
 - Categories
 - Payment Modes

- Locations
- 4. Different date formats were cleaned and standardized.
- 5. Amount values were converted into numeric values.
- 6. Currency symbols and unnecessary commas were removed from amount values.
- 7. Category names were standardized using lookup tables.
- 8. Payment modes were standardized.
- 9. Locations were standardized.
- 10. Missing values were handled with appropriate labels.
- 11. **Check Manually** flags were checked and corrected.
- 12. The cleaned formulas were copied and pasted as **Values** into Clean_Data.

5. Quality Assurance

After the cleaning process, the data was checked using the QA_Check sheet.

Final QA Results

- **Check Manually Flags:** 0
- **Unique Payment Modes:** 4
- **Unique Locations:** 11

The cleaned data was checked before starting the analysis and dashboard creation.

6. Outlier Review

Negative amount values were identified during the data review.

These values were moved to the **Outliers** sheet for further review.

Each outlier was given a reason:

Negative - Review

The suspicious values were documented instead of being deleted from the data without review.

7. PivotTable Analysis

PivotTables were created using the cleaned Clean_Data dataset.

7.1 Category Analysis

The Category PivotTable shows the number of transactions for each category.

7.2 Payment Mode Analysis

The Payment Mode PivotTable shows the number of transactions based on payment method.

The payment modes include:

- UPI
- Card
- Cash
- Bank Transfer

7.3 Location Analysis

The Location PivotTable shows the number of transactions for different locations.

7.4 Transaction Type Analysis

A Transaction Type PivotTable was created to analyze transaction types and amounts.

However, the **Transaction Type chart was intentionally not included in the final Dashboard.**

8. Power Pivot and DAX

Power Pivot was used to create measures for the Dashboard.

Total Amount

Total Amount :=

SUMX(Table1, IFERROR(VALUE(Table1[amount]), 0))

Total Transactions

Total Transactions :=

COUNTROWS(Table1)

Total Income

Total Income :=

SUMX(

Table1,

IF(

Table1[transaction_type] = "Income",

IFERROR(VALUE(Table1[amount]), 0),

```

0
)
)
Total Expense
Total Expense :=
SUMX(
    Table1,
    IF(
        Table1[transaction_type] = "Expense",
        IFERROR(VALUE(Table1[amount]), 0),
        0
    )
)

```

Note: Total Expense was created as a DAX measure, but it was **not used as a KPI card in the final Dashboard**.

9. Final Dashboard

The final Dashboard contains **3 KPI cards** and **3 charts**.

KPI Cards

1. Total Amount

175,791,949

2. Total Transactions

15,836

3. Total Income

125,221,850

The **Total Expense KPI card** was intentionally not included.

Dashboard Charts

The final Dashboard contains:

1. Category Chart

Shows the number of transactions across different categories.

2. Payment Mode Chart

Shows the number of transactions for different payment methods.

3. Location Chart

Shows the number of transactions across different locations.

Final Dashboard Structure

3 KPI Cards:

- Total Amount
- Total Transactions
- Total Income

3 Charts:

- Category
- Payment Mode
- Location

Not included:

- Transaction Type chart
- Total Expense KPI card

10. Tools Used

The following tools and features were used:

- Microsoft Excel
- Excel PivotTables
- Power Pivot
- DAX
- Lookup Tables
- Data Cleaning
- Data Quality Checks
- Outlier Analysis
- Dashboard Creation

- GitHub

11. Key Learning

This project helped in understanding how to:

- Clean real-world financial data
- Handle missing values
- Standardize inconsistent data
- Use lookup tables
- Perform data quality checks
- Identify suspicious transactions
- Create PivotTables
- Use Power Pivot
- Create DAX measures
- Build a finance dashboard
- Prepare a project for GitHub

12. Project Workflow Summary

Raw Data



Unique Value Analysis



Lookup Tables



Data Cleaning



QA Checks



Clean_Data



Outlier Review



PivotTables



Power Pivot / DAX



Dashboard

13. GitHub Repository

Recommended Repository Name

BudgetWise-Personal-Finance-Analysis

Recommended Files

- README.md
- Final Excel workbook
- Raw dataset
- BudgetWise_Finance_Complete_Guide_Updated.pdf

Screenshots are optional and are not required for the project documentation.

14. Final Submission Checklist

Before submitting the project, verify:

- ☒ Raw Data is preserved
- ☒ Lookup tables are completed
- ☒ Cleaning is completed
- ☒ Check Manually Flags = **0**
- ☒ Clean_Data is ready
- ☒ Outliers sheet is created
- ☒ PivotTables are completed
- ☒ Power Pivot measures are created
- ☒ Dashboard has **3 KPI cards**
- ☒ Dashboard has **3 charts**
- ☒ Category chart is included
- ☒ Payment Mode chart is included
- ☒ Location chart is included
- ☒ Transaction Type chart is excluded

- ☑ Total Expense KPI card is excluded
- ☑ README.md is ready
- ☑ Excel workbook is ready for GitHub upload
- ☑ Guide document is ready

15. Conclusion

BudgetWise demonstrates a complete **Excel-based personal finance data analysis workflow**, starting from raw transaction data and ending with a cleaned dataset, quality checks, outlier review, PivotTables, DAX measures, and a simple dashboard.